

Comparing Fractions Without Being Fooled

Directions

- A fraction is a *count* of pieces (the top) and a *size* of piece (the bottom). Comparing only one of the two is the mistake this sheet is about.
- Show the equivalent fractions or the benchmark you used before writing a sign.
- Where a problem asks you to explain, write full sentences on the lines.

1. Fill in the missing numerator: $\frac{3}{4} = \frac{\underline{\quad}}{12}$

Answer: _____

2. Write $<$ or $>$ between the two fractions: $\frac{4}{9}$ $\frac{4}{7}$

Answer: _____

3. Dev writes: “ $\frac{3}{8}$ is greater than $\frac{3}{5}$, because 8 is greater than 5.”

(a) Rewrite $\frac{3}{8}$ with denominator 40 and write its new numerator.

Answer: _____

(b) Rewrite $\frac{3}{5}$ with denominator 40 and write its new numerator.

Answer: _____

(c) Write the correct sign, $<$ or $>$, between $\frac{3}{8}$ and $\frac{3}{5}$.

Answer: _____

(d) Explain what went wrong in Dev’s reasoning.

4. Use one half as a benchmark. Write $<$ or $>$ between the two fractions: $\frac{2}{7}$ $\frac{5}{8}$

Answer: _____

5. Lena writes: “ $\frac{6}{10}$ and $\frac{3}{5}$ must be different amounts, because none of their digits match.”

(a) Write $\frac{6}{10}$ in lowest terms.

Answer: _____

(b) Are the two fractions equivalent? Explain what Lena missed.

6. Sam writes: “ $\frac{5}{6}$ is less than $\frac{7}{12}$, because 5 is less than 7.”

(a) Rewrite $\frac{5}{6}$ with denominator 12 and write its new numerator.

Answer: _____

(b) Write the correct sign, $<$ or $>$, between $\frac{5}{6}$ and $\frac{7}{12}$.

Answer: _____

(c) Explain what went wrong in Sam’s reasoning.

7. Put these three fractions in order from least to greatest: $\frac{3}{5}$, $\frac{7}{10}$, $\frac{2}{3}$

Least _____ Greatest

8. Priya claims: “When two fractions have the same numerator, the one with the bigger denominator is the bigger fraction.”

- (a) Write $<$ or $>$ between $\frac{6}{7}$ and $\frac{6}{13}$.

Answer: _____

- (b) Write $<$ or $>$ between $\frac{5}{12}$ and $\frac{5}{8}$.

Answer: _____

- (c) Write the rule that is actually true for fractions with the same numerator, and explain why it is true.
